



2025

Sql -Database Syllabus



[Type the author name]

Online/ Offline Class

Available

BASIC MYSQL SYLLABUS

Module 1: Introduction to Databases and MySQL

- What is a Database?
 - Types of Databases (Relational, NoSQL, etc.)
 - Introduction to MySQL
 - Installing MySQL and MySQL Workbench
 - MySQL Architecture Overview
 - Connecting to MySQL Server
-

Module 2: Database and Table Operations

- Creating and Selecting Databases
 - Creating Tables (CREATE TABLE)
 - Viewing Table Structure (DESCRIBE, SHOW TABLES)
 - Modifying Tables (ALTER TABLE)
 - Dropping Tables (DROP TABLE)
 - Renaming Tables
-

Module 3: Data Manipulation Language (DML)

- Inserting Data (INSERT INTO)
 - Updating Data (UPDATE)
 - Deleting Data (DELETE)
 - Retrieving Data (SELECT)
 - Using Aliases (AS)
 - Sorting (ORDER BY)
 - Filtering Data (WHERE clause)
-

Module 4: SQL Operators and Clauses

- Arithmetic, Comparison, and Logical Operators
- BETWEEN, IN, LIKE, IS NULL
- DISTINCT and LIMIT clauses

Module 5: Functions in MySQL

- String Functions: CONCAT, LENGTH, SUBSTRING, UPPER, LOWER
 - Numeric Functions: ROUND, CEIL, FLOOR, MOD
 - Date Functions: NOW(), CURDATE(), DATE_FORMAT(), DATEDIFF()
 - Aggregate Functions: COUNT, SUM, AVG, MAX, MIN
-

Module 6: Joins and Relationships

- Understanding Primary and Foreign Keys
 - INNER JOIN
 - LEFT JOIN
 - RIGHT JOIN
 - CROSS JOIN
 - Self Join
 - Using Aliases with Joins
-

Module 7: Data Constraints and Integrity

- Primary Key, Foreign Key, Unique, Not Null
 - Default and Check Constraints
 - Auto Increment
-

Module 8: Subqueries and Nested Queries

- Single Row and Multiple Row Subqueries
 - Using Subqueries with WHERE and FROM clauses
 - EXISTS and NOT EXISTS
-

Module 9: Views and Indexes

- Creating and Managing Views
- Updating Data through Views
- Creating and Dropping Indexes
- Clustered and Non-clustered Indexes

ADVANCED SYLLABUS

Module 10: Advanced Query Techniques

- Complex Joins
- Derived Tables
- UNION and UNION ALL
- GROUP BY with HAVING
- ROLLUP and CUBE

Module 11: Stored Procedures and Functions

- What are Stored Procedures?
- Creating, Altering, and Dropping Procedures
- Using Parameters (IN, OUT, INOUT)
- Creating and Using User-defined Functions
- Error Handling in Procedures

Module 12: Triggers and Events

- What is a Trigger?
- Creating BEFORE and AFTER Triggers
- Using OLD and NEW Keywords
- Creating and Scheduling Events

Module 13: Transactions and Locks

- What is a Transaction?
 - COMMIT, ROLLBACK, and SAVEPOINT
 - ACID Properties
 - Locking Mechanisms (Implicit and Explicit Locks)
 - Isolation Levels
-

Module 14: User Management and Security

- Creating Users and Setting Privileges (GRANT, REVOKE)
 - Roles and Permissions
 - Password Management
 - Backups and Restores
-

Module 15: Performance Optimization

- Query Optimization and EXPLAIN Command
 - Index Tuning
 - Caching and Buffer Optimization
 - Analyzing Slow Queries
-

Module 16: MySQL with Other Tools

- MySQL with Python (mysql.connector / SQLAlchemy)
 - MySQL with PHP / Node.js
 - Importing and Exporting Data (CSV, JSON)
-

Module 17: MySQL Administration

- Understanding MySQL Configuration Files
 - MySQL Server Logs
 - Data Backup and Recovery Techniques
 - Replication Overview
 - Database Migration and Upgrades
-

Module 18: Project / Case Studies

- Create a complete **Database Management Project**:
 - Example: Student Management System / Online Store / Library Management
 - Use Constraints, Joins, Views, Stored Procedures, and Triggers
 - Demonstrate Backup & Restore
-

Optional Certification Topics

- MySQL for Data Analytics (using GROUP BY, JOIN, Aggregate Functions)
 - MySQL in Cloud (AWS RDS, Google Cloud SQL)
 - MySQL for Machine Learning (data preprocessing using SQL)
-

Students Feedback Details

2:01



← Online/Offline Best Ja... 🔍 ⋮

ATES SERVICES **REVIEWS** PHOTOS ABOUT



↗ learning 3



experience 14

kno



Harshit Tyagi

1 review



★★★★★ 6 months ago

There best teacher in this and there teaching method is so good they teach very well our sir was so great they teach things many until student not learn... [more](#)

Reply



Anjali Anjali mahour



★★★★★ 7 months ago

Reply



Tanu Sharma



★★★★★ 7 months ago

Reply



Chat

Saved (2)



2:00



← Online/Offline Best Ja... 🔍 ⋮

ATES SERVICES **REVIEWS** PHOTOS ABOUT



↗ learning 3



experience 14

k



Himani Negi

2 reviews



★★★★★

a day ago

Great experience with Apple Tree Infotech. Mr Umesh Chandra sir is a knowledgeable person and his way of teaching is awesome. He explains... [more](#)

Reply



Rishika Bansal

2 reviews



★★★★★

a day ago

It is a good platform to study about coding and work with it.

Reply



Mujahid Ali

1 review



★★★★★

a day ago

Best training experience and more support all teachers in institute.And produce job any computer



Chat

Saved (2)



← Online/Offline Best Ja... 🔍 ⋮

ATES

SERVICES

REVIEWS

PHOTOS

ABOUT



↗ learning 3

😊 experience 14

kno



Prashant Saraswat

Local Guide · 39 reviews



a day ago

Firstly I would talk about environment of the coaching, looks like every one has known each other from a decade. Sir is too simple and friendly... [more](#)



Reply



Himani Negi

2 reviews



Chat



Saved (2)



← Online/Offline Best Ja... 🔍 ⋮

ATES

SERVICES

REVIEWS

PHOTOS

ABOUT



↗ learning 3

😊 experience 14

kno



Vaishali Tyagi

1 review



a day ago

It has been a very great experience with umesh sir .Training was extremely helpful. Very big thanks to you.



Online/Offline Bes...
a day ago

fotech. (i



Thanku vaishali. Apple tree infotech is a Career development training center. Students is assets for center and Center always try to give b... [more](#)



nandni singh

1 review



a day ago

Apple tree is very good training center. They clear each and every doubts very nicely



Chat



Saved (2)



← Online/Offline Best Ja... 🔍

ATES SERVICES **REVIEWS** PHOTOS ABOUT

📈 learning 3

😊 experience 14

kno

**Aayu Shrivastav**

1 review



★★★★★ a day ago

This place is awesome and teachers also supportive... everyone go there and enhance your skills..

Reply

**ROHIT GUPTA**

3 photos



★★★★★ 2 months ago

Online/Offline Bes...
2 months ago

fotech. (



Thank you rohit write your feedback also

**Aarti Goutam**

★★★★★ 2 months ago



Chat

Saved (2)



← Online/Offline Best Ja... 🔍

ATES SERVICES **REVIEWS** PHOTOS ABOUT

📈 learning 3

😊 experience 14

k

**Himani Negi**

2 reviews



★★★★★ a day ago

Great experience with Apple Tree Infotech. Mr Umesh Chandra sir is a knowledgeable person and his way of teaching is awesome. He explains... [more](#)

Reply

**Rishika Bansal**

2 reviews



★★★★★ a day ago

It is a good platform to study about coding and work with it.

Reply

**Mujahid Ali**

1 review



★★★★★ a day ago

Best training experience and more support all teachers in institute.And produce job any computer



Chat

Saved (2)



2:01



← Online/Offline Best Ja... 🔍 ⋮

ATES SERVICES **REVIEWS** PHOTOS ABOUT

🔍 ↗ learning 3 😊 experience 14 kno



Vani Singh

1 review



★★★★★ 6 months ago

Best and impressive method/procedure of teaching by a lecturer (saurav sir). And exactly what the student accept from the teacher you will get t... [more](#)



Online/Offline Bes...

a day ago

fotech. (i



Thanks bani.



Yaadii Singh



★★★★★ 6 months ago



Reply



techno RK



Chat

Saved (2)



1:58



← Online/Offline Best Ja... 🔍 ⋮

ATES SERVICES **REVIEWS** PHOTOS ABOUT

🔍 ↗ learning 3 😊 experience 14 kno



Tanuj Chaudhary



★★★★★ a day ago



Reply



VISHAL GURJAR

1 review



★★★★★ a day ago

This Coaching is very Supportive nd Helpful for every student , Sir and Staff are very Good in Nature and Has Very Much Knowledge and they guide st... [more](#)



Online/Offline Bes...

a day ago

fotech. (i



Thanku vishal. Center will try always to give best skill and supports to students.



Chat

Saved (2)



2:00



← Online/Offline Best Ja... 🔍 ⋮

ATES

SERVICES

REVIEWS

PHOTOS

ABOUT



📈 learning 3

😊 experience 14

kno

**Mujahid Ali**

1 review



a day ago

Best training experience and more support all teachers in institute. And produce job any computer line courses.

Reply

**Sneha Panchal**

1 review



a day ago

Best place for learning computer language (c,c++, python,java) and also have great learning facility.

Online/Offline Bes...
a day ago

fotech. (i



Thanku sneha.



Chat

Saved (2)



1:58



← Online/Offline Best Ja... 🔍 ⋮

ATES

SERVICES

REVIEWS

PHOTOS

ABOUT



📈 learning 3

😊 experience 14

kno

**Meghna Sharma**

1 review



an hour ago

The Apple tree Infotech was far better than other institutes. The teachers was very nice and very supportive. Their teaching method was so good.

Online/Offline Bes...
just now

fotech. (i



Thanks Meghna. For giving your feedback. I will try to give best service to students.

**Tanuj Chaudhary**

a day ago

Reply



Chat

Saved (2)

